

Experiment 13. Generate a Parse Tree for a Given Sentence

Grammars

Program

```
sentence = input("Enter sentence: ").split()
```

```
if sentence == ["John", "runs"]:
```

```
    print("Parse Tree")
```

```
    print("    S")
```

```
    print("    /\\"")
```

```
    print("  NP  VP")
```

```
    print("  |   |")
```

```
    print("John runs")
```

```
else:
```

```
    print("Sentence not found in grammar")
```

Output

```
# Parse Tree Generation

sentence = input("Enter sentence: ").split()

if sentence == ["John", "runs"]:
    print("Parse Tree")
    print("    S")
    print("    /\\"")
    print("  NP  VP")
    print("  |   |")
    print("John runs")
else:
    print("Sentence not found in grammar")
```

Enter sentence: John runs

Parse Tree

```

      S
     /\
    NP VP
    |  |
  John runs
```

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